← Back to O4 Olympiad geometry

O4 Olympiad geometry: solutions

Read a solution only after a serious attempt. For proof problems, check that your own proof says where the points lie, or uses directed angles so that it does not matter. Notation as in the notes: $a = BC$, $b = CA$, $c = AB$, s the semiperimeter, $[XYZ]$ the area of triangle XYZ.

Level 1

1. Since $AB = AC$, $\angle B = \angle C = \frac{1}{2}(180^\circ - 40^\circ) = 70^\circ$, so $\angle ABD = 35^\circ$. The angle $\angle BDC$ is the exterior angle of triangle ABD at D, so $\angle BDC = \angle A + \angle ABD = 40^\circ + 35^\circ = \mathbf{75}^\circ$.

2. Let $\angle A = x$. Since $AD = BD$, $\angle ABD = x$, and the exterior angle of triangle ABD gives $\angle BDC = 2x$. Since $BD = BC$, $\angle BCD = \angle BDC = 2x$. Since $AB = AC$, $\angle ABC = \angle ACB = 2x$. The angle sum of triangle ABC gives $x + 2x + 2x = 180^\circ$, so $\angle A = \mathbf{36}^\circ$.

3. Inscribed angles on the same chord: $\angle ACB = \angle ADB = 40^\circ$ (chord AB) and $\angle PBC = \angle DBC = 25^\circ$. The angle $\angle APB$ is the exterior angle at P of triangle BPC, so $\angle APB = \angle PBC + \angle PCB = 25^\circ + 40^\circ = \mathbf{65}^\circ$.

4. Let AC and AD meet BE at P and Q. In a diagram the order on line BE is B, P, Q, E. The tip angles at C and E are two angles of triangle PCE (since P lies on CA and on EB), so the exterior angle of that triangle at P is $\angle APE = \angle C + \angle E$, and $\angle APQ = \angle APE$ because Q lies on ray PE. In the same way, triangle QDB has the tip angles at D and B, so $\angle AQP = \angle AQB = \angle B + \angle D$. The angle sum of triangle APQ now gives

$$\angle A + \angle B + \angle C + \angle D + \angle E = \angle A + \angle APQ + \angle AQP = \mathbf{180}^\circ.$$

5. Put the right angle at C with $CA = 3$ and $CB = 4$, and let the square have side x. Above the square is a small right triangle with legs $3 - x$ (along CA) and x (the top side of the square), and it is similar to the whole triangle because its sides are parallel to those of ABC. So $\frac{3-x}{x} = \frac{3}{4}$, which gives $12 - 4x = 3x$ and $x = \mathbf{\frac{12}{7}}$.

6. By the angle bisector theorem, $BD : DC = AB : AC = 6 : 9 = 2 : 3$. So $BD = \frac{2}{5} \cdot 10 = \mathbf{4}$.

7. Power of a point: $AP \cdot PB = CP \cdot PD$, so $24 = 3\,PD$ and $PD = 8$. Then $CD = 3 + 8 = \mathbf{11}$.

8. The tangent length from P is $\sqrt{13^2 - 5^2} = 12$. Power of a point: $PA \cdot PB = 12^2 = 144$, so $PB = 16$ and $AB = PB - PA = 16 - 9 = \mathbf{7}$.

9. $s = 21$, so by Heron the area is $\sqrt{21 \cdot 8 \cdot 7 \cdot 6} = \sqrt{7056} = \mathbf{84}$. Then $r = \frac{84}{21} = \mathbf{4}$ and $R = \frac{13 \cdot 14 \cdot 15}{4 \cdot 84} = \frac{2730}{336} = \mathbf{\frac{65}{8}}$.

Level 2

10. Menelaus in triangle ADC with the line through B (on line DC), E (on AD) and F (on CA), unsigned:

$$\frac{AE}{ED} \cdot \frac{DB}{BC} \cdot \frac{CF}{FA} = 1 \quad \Rightarrow \quad 1 \cdot \frac{1}{3} \cdot \frac{CF}{FA} = 1,$$

so $CF = 3\,FA$ and $AF : FC = \mathbf{1} : \mathbf{3}$. Now Menelaus in triangle BFC with the line through D (on BC), E (on BF) and A (on line FC): here $\frac{CA}{AF} = 4$, so

$$\frac{BD}{DC} \cdot \frac{CA}{AF} \cdot \frac{FE}{EB} = 1 \quad \Rightarrow \quad \frac{1}{2} \cdot 4 \cdot \frac{FE}{EB} = 1,$$

so $BE : EF = \mathbf{2} : \mathbf{1}$.

11. With $a = 7$, $b = 6$, $c = 5$:

$$m_a^2 = \frac{2b^2 + 2c^2 - a^2}{4} = \frac{72 + 50 - 49}{4} = \frac{73}{4},$$

so the median is $\frac{\sqrt{\mathbf{73}}}{\mathbf{2}} \approx 4.27$.

12. $\angle C = 75^\circ$ and $2R = \frac{BC}{\sin A} = \frac{6}{\sqrt{3}/2} = 4\sqrt{3}$. Then

$$AC = 2R \sin B = 4\sqrt{3} \cdot \frac{\sqrt{2}}{2} = \mathbf{2}\sqrt{\mathbf{6}}, \qquad AB = 2R \sin 75^\circ = 4\sqrt{3} \cdot \frac{\sqrt{6} + \sqrt{2}}{4} = \mathbf{3}\sqrt{\mathbf{2}} + \sqrt{\mathbf{6}},$$

using $\sin 75^\circ = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ = \frac{\sqrt{6}+\sqrt{2}}{4}$. The area is

$$\tfrac{1}{2} \cdot AB \cdot AC \cdot \sin 60^\circ = \tfrac{1}{2}(3\sqrt{2} + \sqrt{6})(2\sqrt{6}) \cdot \tfrac{\sqrt{3}}{2} = \tfrac{\sqrt{3}}{2}(6\sqrt{3} + 6) = \mathbf{9} + \mathbf{3}\sqrt{\mathbf{3}}.$$

13. First, line PQ is not parallel to T_1T_2. The line PQ (the radical axis) is perpendicular to the line of centres O_1O_2. If T_1T_2 were also perpendicular to O_1O_2, then the radii O_1T_1 and O_2T_2, which are perpendicular to the tangent, would lie along the line O_1O_2; so T_1 and T_2 would both be the point where O_1O_2 meets the tangent, and the circles would touch there, not meet in two points.

So let PQ meet T_1T_2 at N. Since N lies on the tangent at T_1, its power with respect to the first circle is NT_1^2 (Pythagoras in triangle NT_1O_1); likewise for the second circle. Since N lies on line PQ, both powers also equal $NP \cdot NQ$. So

$$NT_1^2 = NP \cdot NQ = NT_2^2.$$

So $NT_1 = NT_2$ with N on line T_1T_2 and $T_1 \neq T_2$, which means N is the **midpoint of** T_1T_2.

14. Since $\angle BEC = \angle BFC = 90^\circ$, Thales puts E and F on the circle with diameter BC. So B, C, E, F **are concyclic**.

The triangle is acute, so E lies inside segment AC and F inside segment AB, and $BCEF$ is a convex cyclic quadrilateral. Its opposite angles give $\angle FEC = 180^\circ - \angle FBC$. Since A, E, C are collinear with E between them,

$$\angle AEF = 180^\circ - \angle FEC = \angle FBC = \angle ABC.$$

15. From problem 9: area 84, $r = 4$, $R = \frac{65}{8}$. The cosine rule with $a = 14$, $b = 15$, $c = 13$ gives

$$\cos A = \frac{15^2 + 13^2 - 14^2}{2 \cdot 15 \cdot 13} = \frac{198}{390} = \frac{33}{65}, \qquad \sin^2 \tfrac{A}{2} = \frac{1 - \cos A}{2} = \frac{16}{65},$$

so $\sin\frac{A}{2}=\frac{4}{\sqrt{65}}$. Let the incircle touch AB at F. In right triangle AFI, $IF=r$ and $\angle FAI=\frac{A}{2}$, so $AI=\frac{r}{\sin(A/2)}=\frac{4\sqrt{65}}{4}=\sqrt{\mathbf{65}}$.

The inscribed angle $\angle BAM=\frac{A}{2}$ stands on chord BM, so by the sine rule in triangle ABM (which has circumradius R), $BM=2R\sin\frac{A}{2}=\frac{65}{4}\cdot\frac{4}{\sqrt{65}}=\sqrt{65}$. By the incentre/excentre lemma, $MI=MB=\sqrt{\mathbf{65}}$. Since I lies on segment AM and $AI=IM$, I **is the midpoint of** AM. (Numerically $\sqrt{65}\approx 8.06$.)

16. $AB\parallel CM$ and $AB=2\,CM$, so triangles PAB and PCM are similar (alternate angles) with ratio 2. Hence $CP=\frac{1}{3}CA$ and $PM=\frac{1}{3}BM$, so $PM:PB=1:2$. Now $[ABC]=[ACD]=30$. By the ratio lemma, $[BPC]=\frac{1}{3}[BAC]=10$ and $[PCM]=\frac{1}{2}[PCB]=5$. Quadrilateral $APMD$ is triangle ACD with triangle PCM removed, so its area is $30-5=\mathbf{25}$.

17. With tangent lengths (notes, section 5): $BD=BF=s-b$, $CD=CE=s-c$, $AE=AF=s-a$. All three points lie inside the sides, and

$$\frac{BD}{DC}\cdot\frac{CE}{EA}\cdot\frac{AF}{FB}=\frac{s-b}{s-c}\cdot\frac{s-c}{s-a}\cdot\frac{s-a}{s-b}=1.$$

By the converse of Ceva, AD**,** BE**,** CF **are concurrent**. (The common point is called the Gergonne point.)

18. Let the side 7 be opposite angle θ, between the sides 5 and 8. Then $\cos\theta=\frac{25+64-49}{2\cdot5\cdot8}=\frac{40}{80}=\frac{1}{2}$, so $\theta=\mathbf{60}^\circ$. Area $=\frac{1}{2}\cdot5\cdot8\cdot\sin60^\circ=\mathbf{10}\sqrt{\mathbf{3}}$. With $s=10$: $r=\frac{10\sqrt{3}}{10}=\sqrt{\mathbf{3}}$. Sine rule: $R=\frac{7}{2\sin60^\circ}=\frac{7}{\sqrt{3}}=\frac{\mathbf{7}\sqrt{\mathbf{3}}}{\mathbf{3}}$.

19. Put the square at $(0,0)$, $(2,0)$, $(2,2)$, $(0,2)$; the circle passes through $(0,0)$ and $(2,0)$ and touches the side $y=2$. Its centre lies on the perpendicular bisector $x=1$, say at $(1,k)$, and since the circle touches $y=2$ from below, its radius is $2-k$. Then $1+k^2=(2-k)^2=4-4k+k^2$, so $k=\frac{3}{4}$ and the radius is $\frac{\mathbf{5}}{\mathbf{4}}$.

(With power of a point: by symmetry the circle touches the top side at its midpoint T, and the line through T and the midpoint N of the bottom side passes through the centre, so it meets the circle again at the point T' with TT' a diameter. The chords AB and TT' meet at N, so $NA\cdot NB=NT\cdot NT'$, that is, $1\cdot1=2\,NT'$. So $NT'=\frac{1}{2}$, the diameter is $2+\frac{1}{2}=\frac{5}{2}$ and the radius is $\frac{5}{4}$.)

20. The triangle is acute, so H, O, I all lie inside it.

- Let BE, CF be altitudes. In quadrilateral $AFHE$ the angles at E and F are 90°, so $\angle FHE=180^\circ-A$, and $\angle BHC=\angle FHE$ (vertical angles). So $\angle BHC=\mathbf{110}^\circ$.
- Inscribed angle theorem: $\angle BOC=2A=\mathbf{140}^\circ$.
- $\angle BIC=90^\circ+\frac{1}{2}A=\mathbf{125}^\circ$ (notes, Example 1.1).

21. The angle bisector theorem gives $BD:DC=4:6$, so $BD=2$ and $DC=3$. Stewart with $AD=d$:

$$AB^2\cdot DC+AC^2\cdot BD=BC\,(d^2+BD\cdot DC)\quad\Rightarrow\quad16\cdot3+36\cdot2=5(d^2+6),$$

so $120=5d^2+30$, $d^2=18$ and $AD=\mathbf{3}\sqrt{\mathbf{2}}$.

Level 3

22. Answer: $\angle BDE = 30^\circ$**.**

The base angles are $\angle B = \angle C = 80^\circ$. Since $\angle DBC = 60^\circ < 80^\circ$, D lies inside AC and $\angle ABD = 20^\circ$; since $\angle ECB = 50^\circ$, E lies inside AB.

Let G be the point of AC with $\angle GBC = 20^\circ$. Since $20^\circ < 60^\circ$, the order on AC is A, D, G, C.

- Triangle BGC: $\angle GBC = 20^\circ$, $\angle GCB = 80^\circ$, so $\angle BGC = 80^\circ$ and $BG = BC$.
- Triangle BEC: $\angle EBC = 80^\circ$, $\angle ECB = 50^\circ$, so $\angle BEC = 50^\circ$ and $BE = BC$.
- So $BE = BG$, and $\angle EBG = \angle ABC - \angle GBC = 60^\circ$. Hence triangle BEG is equilateral: $GE = GB$ and $\angle BGE = 60^\circ$.
- Triangle BGD: $\angle DBG = 60^\circ - 20^\circ = 40^\circ$ and $\angle BGD = 180^\circ - \angle BGC = 100^\circ$ (as D, G, C are collinear with G between them). So $\angle BDG = 40^\circ$ and $GD = GB$.

So $GD = GE$. Since E lies on segment AB, ray GE lies inside angle $AGB = \angle DGB = 100^\circ$, so $\angle DGE = 100^\circ - 60^\circ = 40^\circ$. Triangle GDE is isosceles with apex angle 40°, so $\angle GDE = 70^\circ$.

Finally, at D: $\angle ADE = 180^\circ - \angle GDE = 110^\circ$ and $\angle ADB = 180^\circ - \angle GDB = 140^\circ$. Since E lies on segment AB, ray DE lies between rays DA and DB, so $\angle BDE = 140^\circ - 110^\circ =$ **30**$^\circ$.

23. The triangle is acute, so H lies inside it, on the same side of BC as A. As in problem 20, $\angle BHC = 180^\circ - A$.

(a) Let H_1 be the reflection of H in BC. Reflection preserves angles, so $\angle BH_1C = \angle BHC = 180^\circ - A$, and H_1 lies on the opposite side of BC from A. So in quadrilateral ABH_1C the opposite angles at A and H_1 sum to 180°, and H_1 **lies on the circumcircle**.

(b) Let H_2 be the reflection of H in M. The diagonals of $BHCH_2$ bisect each other, so it is a parallelogram: $BH_2 \parallel HC$ and $CH_2 \parallel HB$. Also H_2 lies on the opposite side of BC from H, hence from A. With F the foot of the altitude from C, $\angle BCH = \angle BCF = 90^\circ - B$. Alternate angles give $\angle CBH_2 = \angle BCH = 90^\circ - B$, and since A and H_2 lie on opposite sides of BC,

$$\angle ABH_2 = \angle ABC + \angle CBH_2 = B + (90^\circ - B) = 90^\circ.$$

In the same way $\angle ACH_2 = 90^\circ$. Let A' be the point diametrically opposite A. By Thales, $\angle ABA' = \angle ACA' = 90^\circ$. So H_2 and A' both lie on the line through B perpendicular to AB and on the line through C perpendicular to AC. These two lines are not parallel (because AB and AC are not), so they meet in one point: $H_2 = A'$.

(c) In triangle AHA', O is the midpoint of AA' and M is the midpoint of HA' (by (b)). So OM is a midline: $OM = \frac{1}{2}AH$, that is, $AH = 2\,OM$.

24. The triangle is acute, so D, E, F lie inside the sides and H inside the triangle.

Since $\angle BDH = \angle BFH = 90^\circ$, the points B, D, H, F lie on the circle with diameter BH; D and B lie on the same side of line HF (the line CF), so inscribed angles on chord HF give

$$\angle HDF = \angle HBF = \angle EBA = 90^\circ - A$$

(the last step in right triangle ABE). Similarly C, D, H, E lie on the circle with diameter CH, and

$$\angle HDE = \angle HCE = \angle FCA = 90^\circ - A.$$

So $\angle HDF = \angle HDE$. Moreover F and E lie on opposite sides of line AD (one on side AB, the other on side AC), so ray DH is the **internal** bisector of $\angle EDF$. In the same way EH bisects

$\angle DEF$ and FH bisects $\angle DFE$. The internal bisectors of triangle DEF meet at its incentre, and they all pass through H, so H **is the incentre of triangle** DEF.

25. We use directed angles modulo 180°, written $\measuredangle$. They are needed here: depending on which arc P lies on, the feet X, Y, Z can lie on the sides or on their extensions, and an argument with ordinary angles would need several cases.

If P is a vertex, say $P = A$, then $Y = Z = A$ and there is nothing to prove. Otherwise, since $\angle PXC = \angle PYC = 90^\circ$, the points P, X, Y, C lie on the circle with diameter PC; and since $\angle PYA = \angle PZA = 90^\circ$, the points P, Y, Z, A lie on the circle with diameter PA. Then

$$\begin{aligned}
&\measuredangle(YX, YP) = \measuredangle(CX, CP) = \measuredangle(CB, CP) && (P, X, Y, C \text{ concyclic}),\\
&\measuredangle(YZ, YP) = \measuredangle(AZ, AP) = \measuredangle(AB, AP) && (P, Y, Z, A \text{ concyclic}),\\
&\measuredangle(CB, CP) = \measuredangle(AB, AP) && (A, B, C, P \text{ concyclic}).
\end{aligned}$$

So $\measuredangle(YX, YP) = \measuredangle(YZ, YP)$: the lines YX and YZ make the same directed angle with YP and both pass through Y, so they are the same line. Hence X, Y, Z **are collinear**. (If X or Z coincides with Y the statement is trivial.)

26. Let the line through P perpendicular to AB meet AB at F. In triangle APB the angle at P is 90°, so the angles at A and B are acute and F lies inside segment AB. Hence ray PF lies inside angle APB, and the opposite ray PM lies inside the vertical angle CPD; so M lies on segment CD.

In right triangle APB with altitude PF: $\angle APF = 90^\circ - \angle PAF = \angle ABP$. Then

$$\angle MPC = \angle APF = \angle ABP = \angle ABD = \angle ACD = \angle PCM,$$

using vertical angles, and the inscribed angles on chord AD (with B and C on the same arc). So triangle MPC is isosceles: $MP = MC$.

Next, $\angle CPD = 90^\circ$ and M lies inside it, so $\angle MPD = 90^\circ - \angle MPC$. In right triangle CPD, $\angle PDC = 90^\circ - \angle PCD = 90^\circ - \angle MPC$. So $\angle MPD = \angle MDP$ and $MP = MD$. Together, $MC = MP = MD$: M **is the midpoint of** CD.

27. The external bisector at A is parallel to BC exactly when $AB = AC$; the sides are all different, so X, Y, Z exist. None of them is a vertex.

External bisector theorem. X lies outside segment BC and $\frac{XB}{XC} = \frac{AB}{AC}$. *Proof:* the external bisector is perpendicular to the internal bisector, so it does not enter the angle BAC; hence X lies outside segment BC. The external bisector makes the angle $90^\circ - \frac{1}{2}A$ with ray AC and with the extension of BA beyond A, so the angles $\angle XAB$ and $\angle XAC$ are $90^\circ - \frac{1}{2}A$ and $90^\circ + \frac{1}{2}A$ in some order: they add up to 180° and have equal sines. Triangles ABX and ACX have the same height from A, so

$$\frac{XB}{XC} = \frac{[ABX]}{[ACX]} = \frac{\frac{1}{2}\,AB \cdot AX \sin\angle XAB}{\frac{1}{2}\,AC \cdot AX \sin\angle XAC} = \frac{AB}{AC}.$$

So, with signed ratios (negative because each point is outside its segment):

$$\frac{BX}{XC} \cdot \frac{CY}{YA} \cdot \frac{AZ}{ZB} = \left(-\frac{c}{b}\right)\left(-\frac{a}{c}\right)\left(-\frac{b}{a}\right) = -1.$$

By the converse of Menelaus, X**,** Y**,** Z **are collinear**.

28. Put $B=(0,0)$, $C=(14,0)$, $A=(5,12)$; then $AB=13$ and $AC=\sqrt{81+144}=15$. The incentre is the weighted average with weights $a=14$, $b=15$, $c=13$:

$$I=\frac{14\,(5,12)+15\,(0,0)+13\,(14,0)}{42}=\left(\frac{252}{42},\frac{168}{42}\right)=(6,4),$$

which agrees with $r=4$. The circumcentre lies on $x=7$, say $O=(7,y)$, with $OB=R=\frac{65}{8}$: $49+y^2=\frac{4225}{64}$, so $y^2=\frac{1089}{64}$ and $y=\frac{33}{8}$ (positive, since the triangle is acute). Hence

$$OI^2=(7-6)^2+\left(\tfrac{33}{8}-4\right)^2=1+\tfrac{1}{64}=\tfrac{65}{64},\qquad OI=\boxed{\tfrac{\sqrt{65}}{8}}\approx 1.008.$$

Check with Euler's formula $OI^2=R^2-2Rr=\frac{4225}{64}-65=\frac{65}{64}$.

29. Let B_1 and C_1 be the feet of the altitudes from B and C. We use signed products: for a point H and a line through H meeting a circle at U and V, the power of H is $\overline{HU}\cdot\overline{HV}$, negative when H is between U and V.

Since E lies on line AC and $BB_1\perp AC$, we have $\angle BB_1E=90^\circ$ (or $B_1=E$), so B_1 lies on the circle with diameter BE. The line BH is the line BB_1, which meets this circle at B and B_1. So the power of H with respect to the circle with diameter BE is $\overline{HB}\cdot\overline{HB_1}$. In the same way, the power of H with respect to the circle with diameter CF is $\overline{HC}\cdot\overline{HC_1}$.

The points B,C,B_1,C_1 lie on the circle with diameter BC (Thales), and the chords BB_1 and CC_1 meet at H. Power of a point with respect to that circle gives $\overline{HB}\cdot\overline{HB_1}=\overline{HC}\cdot\overline{HC_1}$. So H **has equal power** with respect to both circles, and lies on their radical axis (if the circles are different). When the circles meet in two points, the radical axis is the line through those points, so **that line passes through** H.

30. Answer: $\frac{1}{7}$.

Let $X=AD\cap BE$, $Y=BE\cap CF$, $Z=CF\cap AD$, and $S=[ABC]$.

Menelaus in triangle ADC with the line through B, X, E (unsigned):

$$\frac{AX}{XD}\cdot\frac{DB}{BC}\cdot\frac{CE}{EA}=1\quad\Rightarrow\quad\frac{AX}{XD}\cdot\frac{1}{3}\cdot\frac{1}{2}=1\quad\Rightarrow\quad AX:XD=6:1.$$

Menelaus in triangle ABD with the line through C, Z, F:

$$\frac{AZ}{ZD}\cdot\frac{DC}{CB}\cdot\frac{BF}{FA}=1\quad\Rightarrow\quad\frac{AZ}{ZD}\cdot\frac{2}{3}\cdot 2=1\quad\Rightarrow\quad AZ:ZD=3:4.$$

So on AD the order is A,Z,X,D with $AZ:ZX:XD=3:3:1$. By the ratio lemma, $[ABD]=\frac{1}{3}S$ and

$$[ABX]=\frac{AX}{AD}\,[ABD]=\frac{6}{7}\cdot\frac{1}{3}S=\frac{2}{7}S.$$

The conditions are unchanged when we rename $A\to B\to C\to A$; this renames $D\to E\to F\to D$ and $X\to Y\to Z\to X$. So also $[BCY]=[CAZ]=\frac{2}{7}S$. The triangle ABC is cut by the three cevians into the triangles ABX, BCY, CAZ and XYZ (draw it: ABX is bounded by AB, part of BE and part of AD). Therefore

$$[XYZ] = S - 3 \cdot \frac{2}{7} S = \tfrac{1}{7}\, S.$$